



P3Q Corona Kind: Module-LWE Threshold Verify on the Unified Precompile Slot

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Abstract

LP-218 introduces P3Q at EVM slot 0x012205 as the PQ-verifier family for rollup-batch authentication, specifying the unified kind-byte dispatch ABI and the first verifier kind 0x01 (Pulsar / FIPS-204 ML-DSA-65 threshold). This paper specifies *kind* 0x02 *Corona*: a Module-LWE threshold verifier built on the two-round Corona construction (eprint 2024/1113). The shared P3Q framework — the unified slot, the kind-byte dispatch ABI, and the **P3QVerifier** registration interface — is specified in LP-219; this paper gives Corona’s wire layout and its gas calibration (5 000 000 gas). Corona reduces to Module-LWE / ring-structured lattice hardness, independent of Pulsar’s Module-LWE (LP-223) and Magnetar’s hash-collision (LP-222) assumptions, and is the Module-LWE parameter-diversity leg of the LP-217 PQ-strict posture. We also place Corona in the DEX rollup settlement pipeline: per-market fill batches roll up to the Z-Chain and are verified on a two-rung ladder — a kind 0x01 ML-DSA-65 attestation rung (measured ~ 4714 verifies/s) and a **starkfri** STARK/FRI validity rung (LP-221, measured ~ 986 /s on a toy AIR) — with Corona as the optional Module-LWE second leg for value-bearing markets and bundled $K \times V \times$ markets settlement. GPU dispatch is projected, not measured; the on-chain path reserves Corona’s dispatch entry and returns **abiFalse** via **ErrKindNotImplemented** until its threshold library lands.

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1 Introduction

P3Q (LP-219) is the unified-slot post-quantum verifier framework at EVM precompile slot 0x012205: one slot, one ABI, and a leading `kind` byte that dispatches into a per-kind threshold-signature verifier core. This paper specifies *kind 0x02 Corona*, the Module-LWE member of the P3Q family.

For the slot allocation, the kind-byte dispatch ABI, the `P3QVerifier` Go interface that registers each kind, and the cross-family composition into LP-217 cert profile modes, see LP-219. The sibling kinds are specified in LP-223 (Pulsar / rank- k Module-LWE) and LP-222 (Magnetar / hash-based FIPS-205 SLH-DSA).

Corona reduces to the hardness of Module-LWE / ring-structured lattices — built on the two-round Corona threshold construction (eprint 2024/1113) — which is independent of the Module-LWE and hash-collision assumptions its sibling kinds rest on. It is the Module-LWE parameter-diversity leg of the LP-217 PQ-strict posture and is calibrated at 5 000 000 gas, the cost cliff that buys cross-family lattice diversity. Performance numbers are tagged **measured** or **projected**; the on-chain path reserves Corona’s dispatch entry and returns `abiFalse` via `ErrKindNotImplemented` until its threshold library lands, and GPU dispatch is projected, not measured.

2 Corona Kind (kind = 0x02)

2.1 Slot and ABI

0x012205 with leading `kind=0x02`. Same physical precompile address as Pulsar (kind 0x01) and Magnetar (kind 0x03); dispatch resolves to the Corona `P3QVerifier` implementation.

```
// EVM precompile at slot 0x012205, kind = 0x02
bool ok = P3Q.verify(
    0x02,                // Corona kind
    coronaSig,           // serialized Corona threshold sig
    messageHash,         // H(prev_root || new_root || batch_hash)
    groupPubKey          // Corona group public key
);
```

2.2 Underlying scheme

Corona [2] is the Lux deployment of a two-round lattice threshold signature scheme building on the Ringtail line [1]. Hardness reduces to Module-LWE over the ring $R_q = \mathbb{Z}_q[X]/(X^{256} + 1)$ at the Ringtail parameter set (module matrix $A \in R_q^{8 \times 7}$, 48-bit prime q), via the standard two-round distributed signing protocol.

Dealerless key generation — the permissionless leg. Unlike Pulsar — whose byte-equality to FIPS 204 *forces* a trusted-dealer key generation, because a dealerless sum of N small-norm FIPS secrets escapes the ML-DSA S_η bound (LP-073 [3]) — Corona is *natively dealerless*. Key generation runs a Pedersen-style verifiable-secret-sharing DKG over R_q (the `dkg2` package: commitment $\mathbf{C} = \mathbf{A} \cdot \text{NTT}(c) + \mathbf{B} \cdot \text{NTT}(r)$, commitment broadcast, a digest cross-check round, and verification with signed-complaint identifiable abort), so no single party ever holds the group secret. As of `corona v0.8.0` this is the production keygen default (`keyera.Bootstrap`). Signing likewise never reconstructs the key: the combiner sums partial responses ($\mathbf{z} = \sum_j \mathbf{z}_j$, then $\mathbf{A}\mathbf{z}$), and the master secret is never formed. Consequently, in the Quasar dual-PQ AND-mode certificate (LP-020), **Corona is the leg that carries the permissionless-public guarantee** a sovereign validator set requires — the property Pulsar’s trusted-dealer keygen cannot provide.

Corona’s signature is *not* byte-equal to any NIST standard. Both Corona and Pulsar (FIPS-204 ML-DSA-65) are Module-LWE schemes; they differ in *parameter regime and codebase*, not

in hardness family. Corona uses the Ringtail-derived parameter set (~ 33 KB signatures, NIST Cat-1) with Module-SIS polynomial commitments; Pulsar uses the ML-DSA-65 tuning (~ 3.3 KB, Cat-3, byte-equal to FIPS-204). Pairing them gives parameter- and implementation-level diversity — a break specific to one parameter set or codebase need not hit the other — but *not* family-disjoint hardness: a structural break of Module-LWE would threaten both. Cross-family independence on the signature lane comes only from the hash-based Magnetar kind (kind 0x03, SLH-DSA).

2.3 Wire format

Offset	Size	Field
0	1	kind = 0x02
1	1	mode = 0x01 (reserved Corona parameter set)
2	4	sigLen (BE32)
6	sigLen	coronaSig ($\sim 33\,052$ B)
6+sigLen	4	pkLen (BE32)
10+sigLen	pkLen	groupPubKey
10+sigLen+pkLen	32	messageHash

Table 1: Corona-kind wire format. Sig size is the dominant cost (~ 33 KB); the rest mirrors the Pulsar layout to keep parsing uniform across kinds.

The mode byte is reserved at 0x01; future Corona parameter set additions (a higher-security Module-LWE variant) get new mode bytes without changing the wire-format skeleton.

2.4 Verifier core

The Go reference implementation is planned at `lux/standard/precompiles/p3q_family/corona.go` and will wrap `luxfi/corona/sign.Verify`. The core is bespoke – not FIPS-204 byte-equal – because Corona is a separate Module-LWE construction (Ringtail-derived), not a NIST standard.

Status as of 2026-06-04: the verifier core is not yet wired into the P3Q dispatch table. The reference `p3q.Run` routes `kind=0x02` to `abiFalse` via the `ErrKindNotImplemented` path:

```
case KindCorona, KindMagnetar:
    // Reserved kinds -- verifier not yet wired. Return abiFalse
    // ...
    return abiFalse(), remaining, nil
```

Solidity callers MAY emit `P3Q.verify(0x02, ...)` from network genesis; the call charges the per-kind gas (see below) and returns `false` until the verifier lands. This is the LP-218 “no revert on cryptographic failure” contract: `false` on the return word is the universal signal for “not verified for any reason,” and reserved-but-not-wired joins “malformed input” and “FIPS reject” under that umbrella.

2.5 Gas cost

5 000 000 gas – $\sim 100\times$ the Pulsar kind’s 50 000. Returned by the Corona `P3QVerifier.GasCost()` entry. Why the disparity?

1. **CPU verify cost is heavier.** The Corona verifier requires NTT inverse over the Module-LWE ring + polynomial multiplication mod q over the larger Ringtail parameter set (~ 33 KB sig vs Pulsar’s 3.3 KB).
2. **Calibration uses the CPU verify cost as the baseline.** LP-218’s calibration discipline assumes a CPU-class verifier – validators are not required to have Blackwell GPUs. Under

that assumption, Corona’s effective per-validator verify cost is the parity audit’s measured ~ 1.6 ms CPU, which is approximately $20\times$ the Pulsar CPU verify time ($\sim 80\mu\text{s}$). The gas charge is rounded up to $100\times$ Pulsar’s for conservative accounting: 5 M gas.

3. **Block-budget headroom.** At 12 M block gas and 5 M per Corona verify, the limit is ~ 2 Corona verifies per block – this is the bottleneck for very high- density PQ-strict rollup throughput. The cost cliff is intentional; the LP-217 mode ladder uses it to price cross-family insurance correctly.

Operators with all-Blackwell validator sets may petition for a **Blackwell-discounted Corona gas charge** in a future amendment; LP-220 does not codify it. The conservative CPU-baseline gas charge applies until validator-hardware uniformity is established. This is the same calibration discipline LP-218 uses for the Pulsar kind.

2.6 Performance

Metric	M1 Max	Spark	Source
Per-validator Corona sign at $N=64$	~ 30 ms	~ 30 ms	matrix bench (measured) [6]
CPU verify (single sig)	~ 1.6 ms	~ 1.6 ms	parity audit (measured)
Blackwell sm_120 verify	—	~ 15 ms	projected (LP-203 kernel) [4]
Blackwell sm_120 batched ($N=64$)	—	~ 50 ms aggregate	projected

Table 2: Corona kind performance. Sign and CPU-verify times are measured per the parity audit; Blackwell verify timings remain projections until the GPU dispatch boundary closes (see LP-223).

2.6.1 Scaling exponent

The parity audit measured Corona-leg signing cost growing super-linearly in N . The measured exponent across $N \in \{16, 32, 64\}$ is approximately $T(N) \approx c \cdot N^{1.5-1.7}$, depending on the host platform. This is not an $O(N)$ scheme and should not be described as such in downstream documents; the Module-LWE distributed signing protocol’s full-rank check and Gaussian-hash PRF carry costs that grow faster than linearly in the participant count. The measured exponent is the operative number for sequencer capacity planning, not the naive-amortization figure.

2.7 Use case

A rollup that wants **intra-lattice parameter and implementation diversity** configures its sequencer to sign each batch with both a Pulsar threshold sig AND a Corona threshold sig — two Module-LWE schemes at distinct parameter sets and codebases. The rollup contract on Z-Chain calls `P3Q.verify(0x01, ...)` and `P3Q.verify(0x02, ...)` at the same precompile address `0x012205` in sequence inside its `validateBatch` function; both must return `true` for the batch to be accepted.

A break specific to Pulsar’s parameter set or codebase need not compromise Corona’s, so the rollup retains a valid PQ leg under an implementation- or parameter-localized failure. This is *not* family-disjoint hardness: a structural break of Module-LWE itself would threaten both legs. For cross-family insurance the rollup must add the hash-based Magnetar kind (`P3Q.verify(0x03, ...)`). The cost is $\sim 5\,050\,000$ gas per batch (Pulsar $50\,000$ + Corona $5\,000\,000$). Acceptable for value-bearing rollups (custody, treasury, regulated securities); prohibitive for high-volume rollups (gaming, social).

This is the LP-217 PQ-strict posture made operationally callable.

3 Application: DEX Rollup Settlement to Z-Chain

The DEX matching path (Lux Lightspeed DEX [7]) fills orders at hundreds of millions per second per GPU on the D-Chain, far above any single chain’s finality rate. To anchor that throughput trustlessly, per-market fill batches roll up to the Z-Chain and are verified there through the P3Q family at slot 0x012205. This section places Corona (kind 0x02) in that settlement pipeline; the measured verify rates belong to the sibling kinds, and Corona is the optional Module-LWE parameter-diversity leg.

3.1 Two verification rungs

DEX rollup settlement uses a two-rung ladder, both callable at the same P3Q slot:

1. **Attestation rung (venue-trust).** The sequencer signs each per-market batch with a P3Q kind 0x01 (Pulsar / FIPS-204 ML-DSA-65) threshold signature. The Z-Chain contract calls `P3Q.verify(0x01, ...)` to confirm the batch was produced by the venue’s threshold quorum. This rung trusts the venue’s signer set but is cheap: the ML-DSA-65 verify path is measured at ~ 4714 verifies per second.
2. **Validity rung (trustless).** For trust-minimized settlement the sequencer additionally produces a STARK/FRI validity proof of correct matching, verified by the `starkfri` verifier (LP-221 [5]) — a cSHAKE256 / Goldilocks PQ Plonky3-fork verifier. This rung proves the fills are correct without trusting the signer set. The current verifier is measured at ~ 986 proofs per second against a **toy AIR**; this figure is the operative number and is explicitly not a production-circuit measurement.

The attestation rung is fast and venue-trusting; the validity rung is slower and trustless. A venue chooses the rung (or runs both) per its trust requirements.

3.2 Where Corona fits

Corona (kind 0x02) is the *Module-LWE cross-family insurance leg* for the attestation rung, not a separate settlement mechanism. A value-bearing DEX rollup that wants independence from the Module-LWE assumption underlying ML-DSA-65 signs each batch with *both* a Pulsar (kind 0x01) and a Corona (kind 0x02) threshold signature and requires both `P3Q.verify` calls to return `true`, exactly as in Section 2’s use case. A Module-LWE break that compromises the Pulsar attestation does not compromise the Corona attestation; the rollup’s settlement remains cryptographically anchored.

The cost discipline of Section 2 applies unchanged: Corona’s 5 000 000 gas makes the second leg appropriate for value-bearing markets (custody, treasury, regulated-securities venues) and prohibitive for high-volume markets, which settle on the kind 0x01 attestation rung alone (optionally backed by the starkfri validity rung). As noted throughout this paper, Corona’s GPU dispatch is projected and its verifier core is not yet wired; the DEX settlement path that needs a second lattice family today uses the measured kind 0x01 rung, with Corona reserved at its dispatch entry until its threshold library lands.

3.3 Bundled settlement: $K \times V \times$ markets

Per-fill, per-venue verification does not scale: at the attestation-rung’s ~ 4714 verifies per second, a chain that verified one signature per fill would cap settlement far below the matcher’s fill rate. The settlement model therefore bundles. A single Z-Chain settlement object aggregates K signer attestations across V venues across many markets into one verification event:

- **K signers.** The threshold quorum per venue contributes K partial signatures aggregated into one threshold signature that verifies in a single `P3Q.verify` call.
- **V venues.** Independent venues’ batches are aggregated into one settlement object rather than verified one venue at a time.

- **Many markets.** Each venue’s per-market arenas (Lux Lightspeed DEX) contribute their batch roots to the bundle, so a single verification covers many books.

Because one verification amortizes over $K \times V \times$ markets fills, the effective settled-fill rate at fleet scale is **projected** to exceed 10^9 fills per second of finality — this is a fleet-aggregation projection, not a single-verifier measurement. The per-verifier measured rates ($\sim 4714/s$ attestation, $\sim 986/s$ validity on a toy AIR) are the operative numbers; the $>10^9$ figure is what those rates compose to once the bundle factor $K \times V \times$ markets is applied across a multi-node Z-Chain settlement fleet. No single P3Q verifier reaches that rate.

References

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